

Theory of specialization of the fundamental group

SGA 1, Exposé X

English draft

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About this draft

This is an unofficial English translation of Exposé X, “Théorie de la spécialisation du groupe fondamental”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ X

Theory of specialization of the fundamental group

In the present exposé, we confine ourselves to the study of the fundamental group of the geometric fibers in a *proper* morphism, i.e. of the fundamental group of a variable *proper* algebraic scheme. In a later exposé, we shall generalize the technique employed to *tamely ramified* étale coverings “at infinity”. This will give us for example a solution of the “Three-point problem” in the case of Galois coverings of order prime to the characteristic (i.e. a determination of the Galois coverings of the line $\mathbf{P}_k^1$, ramified at most at three given points and tamely ramified at these points), and of its obvious variants.

1. The homotopy exact sequence for a proper and separable morphism

DEFINITION 1.1. A prescheme X over a field k is said to be *separable*, or *separable over k* , if for every extension K of k , $X \otimes_k K$ is reduced. If $f: X \rightarrow Y$ is a morphism of preschemes, one says that f is *separable*, or that X is *separable over Y* , if X is flat over Y and if for every $y \in Y$, the fiber $X \otimes_Y k(y)$ is separable over $k(y)$.

If X is a prescheme over a field k , to say that it is separable also means that it is *reduced*, and that the fields $k(x)$, for x a generic point of an irreducible component of X , are separable extensions of k . If k is perfect, it therefore amounts to the same to say that X is reduced. Let us note that if X is separable over Y , then for every change of base $Y' \rightarrow Y$, $X' = X \times_Y Y'$ is separable over Y' . One can also prove, under suitable finiteness hypotheses, that the composite of two separable morphisms is a separable morphism. We shall need this only in the following form: *if X is separable over Y , and X' étale over X , X' is separable over Y* . This is indeed an immediate consequence of the definitions and I 9.2. Moreover, the hypothesis “separable morphism” will serve us through the following proposition:

PROPOSITION 1.2. *Let $f: X \rightarrow Y$ be a proper and separable morphism, with Y locally noetherian, and consider its Stein factorization $X \xrightarrow{f'} Y' \rightarrow Y$, (where $f'_*(\mathcal{O}_X) = \mathcal{O}_{Y'}$, Y' being finite over Y and isomorphic to the spectrum of the algebra $f_*(\mathcal{O}_X)$). Then Y' is an étale covering of Y .*

This proposition will appear in (EGA III 7)¹. Let us indicate the principle of the proof. One easily reduces to the case where Y is the spectrum of a complete local ring A , and, making moreover a suitable finite flat extension of the latter (corresponding to a suitable residual extension), one can assume that the connected components of the fiber of the closed point y are geometrically connected, which also means that $H^0(X_y, \mathcal{O}_{X_y})$ decomposes into a product of fields identical to $k = k(y)$. Assuming then X connected, which is permissible, one will have $H^0(X_y, \mathcal{O}_{X_y}) = k$, hence the homomorphism $A \rightarrow H^0(X_y, \mathcal{O}_{X_y})$ is *surjective*. One concludes from this by a general proposition (of Künneth type) that $f_*(\mathcal{O}_X)$ is defined by a module B over A which is free over A , and that $B/\mathfrak{m}B \rightarrow H^0(X_y, \mathcal{O}_{X_y}) = k$ is bijective. Hence in the present case B is an étale algebra over A , which completes the proof.

THEOREM 1.3. *Let $f: X \rightarrow Y$ be a proper and separable morphism, with Y locally noetherian and connected, and assume $f_*(\mathcal{O}_X) = \mathcal{O}_Y$ (which implies that the fibers of X over Y are geometrically connected, and conversely thanks to 1.2). Let y be a point of Y , $\overline{k(y)}$ an algebraic closure of $k(y)$, $\overline{X}_y = X_y \otimes_{k(y)} \overline{k(y)}$. Finally, let X' be a connected étale*

¹Cf. EGA III 7.8.10 (i)

covering of X , and $\overline{X}'_y = X'_y \otimes_{k(y)} \overline{k(y)}$. For there to exist an étale covering Y' of Y and an X -isomorphism $X' \xrightarrow{\sim} X \times_Y Y'$, it is necessary and sufficient that $\overline{X}'_y$ admit a section over $\overline{X}_y$.

Setting $Y' = \text{Spec}(h_*(\mathcal{O}_{X'}))$ (where $h: X' \rightarrow Y$ is the composite morphism $X' \rightarrow X \rightarrow Y$), it suffices to prove that the canonical Y -morphism

$$X' \longrightarrow X \times_Y Y'$$

is an *isomorphism*, and that Y' is étale over Y . Now we already know by 1.2 that Y' is étale over Y , hence $X \times_Y Y'$ is étale over X , hence the morphism $X' \rightarrow X \times_Y Y'$ is also étale (I 4.8). Moreover, Y' is connected as the image of X' which is so, hence $X \times_Y Y'$ is connected since X has connected fibers over Y (IX 3.4 and V 6.9 (iii)). Hence to prove that $X' \rightarrow X \times_Y Y'$ is an isomorphism, it suffices to see that its degree of projection at *one* point of $X \times_Y Y'$ is equal to 1. Now this follows easily from the hypothesis that $\overline{X}'_y$ admits a section over $\overline{X}_y$, either by use of IX 6.6, or more simply by noting that it suffices to prove the existence of such a point in $X \times_Y Y'$ after change of base $\text{Spec}(\overline{k(y)}) \rightarrow Y$, where this is evident. This completes the proof of 1.3.

Taking into account IX 3.4 and the dictionary V 6.9 and V 6.11, one can put 1.3 in the following equivalent form

COROLLARY 1.4. *With the preceding notation for $f: X \rightarrow Y$ and $\overline{X}_y$, let $\bar{a}$ be a geometric point of $\overline{X}_y$, a its image in X and b its image in Y . Then the following sequence of group homomorphisms is exact:*

$$\pi_1(\overline{X}_y, \bar{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

REMARKS 1.5. One will note that the proof of 1.3 makes essential use of 1.2 and thereby of the “first comparison theorem” in formal algebraic geometry. On the other hand, the descent theory of Exposé IX has intervened only through IX 3.4, of which a direct proof in the case of a *proper* morphism $f: X \rightarrow Y$ such that $f_*(\mathcal{O}_X) = \mathcal{O}_Y$ is easy. Indeed, let Y' be étale over Y and assume that $X' = X \times_Y Y'$ is the disjoint sum of two nonempty open sets; let us prove that the same holds for Y' . Indeed, one will have $Y' = \text{Spec}(\mathcal{A})$, hence $X' = \text{Spec}(\mathcal{B})$ with $\mathcal{B} = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_X$, and the decomposition of X' as a direct sum corresponds to a decomposition of $\mathcal{B}$ as a product of two nonzero Algebras $\mathcal{B}_1$ and $\mathcal{B}_2$. Since $f_*(\mathcal{O}_X) = \mathcal{O}_Y$ one easily concludes $f_*(\mathcal{B}) = \mathcal{A}$, hence $\mathcal{A}$ will be the sum of two Algebras (likewise nonzero, since their unit sections are nonzero) $f_*(\mathcal{B}_1)$ and $f_*(\mathcal{B}_2)$, qed.

1.6. Let us again assume that f is proper and separable, but let us no longer make any hypothesis on $f_*(\mathcal{O}_X)$, which will correspond to a well-determined étale covering Y' of Y , moreover pointed over b by the image b' of a . Applying then 1.4 to the canonical morphism $X \rightarrow Y'$, and assuming f surjective, the exact sequence 1.4 is replaced by the following one, analogue of the homotopy exact sequence of fiber spaces in algebraic topology:

$$\pi_1(\overline{X}_y, \bar{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow \pi_0(\overline{X}_y, \bar{a}) \longrightarrow \pi_0(X, a) \longrightarrow \pi_0(Y, b) \longrightarrow e.$$

Of course, in 1.4 one cannot in general assert that the homomorphism $\pi_1(\overline{X}_y, \bar{a}) \rightarrow \pi_1(X, a)$ is injective; in algebraic topology, its kernel is the image of $\pi_2(Y, b)$, and there would be occasion in algebraic geometry as well to introduce homotopy groups in all dimensions, and the complete homotopy exact sequence for a proper morphism satisfying suitable hypotheses (for example that of being a smooth morphism). One does not have at present any result in this direction, with the exception of a reasonable (if not definitive) definition of the higher homotopy groups.

COROLLARY 1.7. *Let k be an algebraically closed field, X and Y two connected preschemes over k ; one assumes X proper over k and Y locally noetherian. Let a be a geometric point of X , b a geometric point of Y with values in the same algebraically closed extension K of k ; consider the geometric point $c = (a, b)$ of $X \times_k Y$, and the homomorphism $\pi_1(X \times_k Y, c) \rightarrow$*

$\pi_1(X, a) \times \pi_1(Y, b)$ deduced from the homomorphisms on the fundamental groups associated with the two projections $X \times_k Y \rightarrow X$ and $X \times_k Y \rightarrow Y$. The preceding homomorphism is an isomorphism.

Let us first assume $K = k$. Let us set $Z = X \times_k Y$, consider the projection $f: Z \rightarrow Y$ and the locality y of the geometric point b of Y , and let us apply to the situation the result 1.4. One will note for this that, at the cost of passing to X_{red} (which does not change the fundamental groups under consideration), one can already assume X reduced, hence separable over k , hence Z is separable over k , and evidently with geometrically connected fibers (since X is connected). The geometric fiber of Z at b is canonically isomorphic to $X \otimes_k K = X$. On the other hand, since the composite of the morphisms $X \rightarrow Z \rightarrow X$ is the identity, one finds that $\pi_1(X, a) \rightarrow \pi_1(Z, c)$ is injective, and 1.4 gives us an exact sequence:

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Z, c) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

On the other hand, one has the canonical exact sequence:

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b) \longrightarrow \pi_1(Y, b) \longrightarrow e,$$

where the two homomorphisms written are the canonical injection and the canonical projection. Finally, one has a homomorphism from the first exact sequence into the second, by means of the identity morphisms on the extreme terms, and the canonical homomorphism $\pi_1(Z, c) \rightarrow \pi_1(X, a) \times \pi_1(Y, b)$ for the middle terms. The commutativity of the diagram thus obtained is verified trivially. Since the homomorphisms on the extreme terms are isomorphisms, the same holds for the middle terms, which proves 1.7 in this case.

When one no longer assumes $K = k$, one finds only an isomorphism

$$\pi_1(Z, c) \longrightarrow \pi_1(X \otimes_k K, a) \times \pi_1(Y, b),$$

and 1.7 is then equivalent to the following particular case:

COROLLARY 1.8. *Let X be a proper and connected scheme over an algebraically closed field k , k' an algebraically closed extension of k , a' a geometric point of $X \otimes_k k'$ and a its image in X . Then the canonical homomorphism $\pi_1(X \otimes_k k', a') \rightarrow \pi_1(X, a)$ is an isomorphism.*

The fact that this homomorphism is surjective is equivalent to saying that if X' is a connected étale covering of X , then $X' \otimes_k k'$ is also connected, and follows immediately from the fact that k is algebraically closed; it is also a particular case of IX 3.4. The hypothesis of properness on X has not yet been used. That said, to say that the homomorphism under consideration is injective also means this: *every étale covering of $X \otimes_k k'$ is isomorphic to the inverse image of an étale covering of X* . It is essentially a matter of sorites that one can find a sub- k -algebra A of k' , of finite type over k , and an étale covering of $X \otimes_k A$ whose inverse image on $X \otimes_k k'$ is isomorphic to the given covering. Let then $Y = \text{Spec}(A)$, which is an integral k -scheme of finite type, hence has rational points over k . Let us then apply 1.7 to the fundamental group of $X \times Y$ at a point (a, b) rational over k : one finds that every connected étale covering of $X \times Y$ is isomorphic to a quotient of a covering $X' \times Y'$, where X', Y' are Galois étale coverings of X and Y with groups G, G' , by a subgroup H of $G \times G'$. This implies that the inverse image of this covering of $X \times Y$ on $X \times Y'$ is isomorphic to a covering of the form $X'_1 \times Y'$, where X'_1 is an étale covering of X . If then L is the field of functions of Y , equal to the field of fractions of A in k' , the étale covering of $X \otimes_k L$ induced by the given covering of $X \times_k Y$ is such that there exists a finite separable extension L' of L such that the inverse image of the said covering on $X \otimes_k L'$ is isomorphic to $X'_1 \otimes_k L'$. Now, k' being algebraically closed, one can assume that the extension L' of L is contained in k' . This proves that the given étale covering of $X \otimes_k k'$ is isomorphic to $X'_1 \otimes_k k'$, qed.

The explicit form noted in passing for the étale coverings of a product $X \times_k Y$ immediately implies the following result:

COROLLARY 1.9. *Let k be an algebraically closed field, X and Y two locally noetherian preschemes over k , $Z = X \times_k Y$ their product, Z' an étale covering of Z . For every point $y \in Y$*

rational over k , let $i_y: \text{Spec}(k) \rightarrow Y$ be the canonically associated morphism, $j_y = \text{id}_X \times_k i_y$ the corresponding morphism $X \rightarrow Z$. Finally, let X'_y be the étale covering of X inverse image of Z' by j_y . One assumes Y connected, and X or Y proper over k . Then the coverings X'_y of X are all isomorphic.

Figuratively, one can say that a family of étale coverings of X , parametrized by a connected prescheme Y , is constant if X or the parameter prescheme Y is proper over k .

REMARKS 1.10. Corollaries 1.7 to 1.9 are due to Lang-Serre [2] in the case of normal algebraic schemes. (Their work was the initial motivation for the theory of the fundamental group developed in this Seminar). As these authors have remarked, these results become false when one drops the hypothesis of properness, at least in characteristic $p > 0$. Taking for example for X the affine line $X = \text{Spec}(k[t])$, it is not difficult to see that the coverings of X , parametrized by the affine line $Y = \text{Spec}(k[s])$, defined by the equations

$$x^p - x = st,$$

are étale and pairwise non-isomorphic. This invalidates 1.9 and a fortiori 1.7, and one sees similarly that if s is considered as an element transcendental over k in an algebraically closed extension K of k , one finds an étale covering X' of $X \otimes_k K$ which does not come from an étale covering of X .

2. Application of the existence theorem for sheaves: semicontinuity theorem for the fundamental groups of the fibers of a proper and separable morphism

THEOREM 2.1. *Let Y be the spectrum of a complete noetherian local ring, with residue field k , X a proper Y -scheme, $X_0 = X \otimes_A k$, a_0 a geometric point of X_0 and a the corresponding geometric point of X . Then the canonical homomorphism $\pi_1(X_0, a_0) \rightarrow \pi_1(X, a)$ is an isomorphism.*

This is only a translation, into the language of the fundamental group, of the result recalled in IX 1.10. It is here that the existence theorem for sheaves in formal algebraic geometry enters in an essential way into the theory of the fundamental group.

Let us now introduce an algebraic closure $\bar{k}$ of the residue field k , and the geometric fiber $\bar{X}_0 = X_0 \otimes_k \bar{k}$. We thus have the exact sequence (IX 6.1)

$$e \longrightarrow \pi_1(\bar{X}_0, \bar{a}) \longrightarrow \pi_1(X_0, a_0) \longrightarrow \pi_1(k, \bar{k}) \longrightarrow e,$$

on the other hand we have the isomorphism 2.1 and the analogous, more elementary, isomorphism

$$\pi_1(k, \bar{k}) \longrightarrow \pi_1(Y, b),$$

where b is the image of a in Y . We thus find:

COROLLARY 2.2. *With the preceding notation, suppose $\bar{X}_0 = X_0 \otimes_k \bar{k}$ connected, and let $\bar{a}_0$ be a geometric point of $\bar{X}_0$, a_0 its image in X , b_0 its image in Y . Then the following sequence of canonical homomorphisms*

$$e \longrightarrow \pi_1(\bar{X}_0, \bar{a}_0) \longrightarrow \pi_1(X, a_0) \longrightarrow \pi_1(Y, b_0) \longrightarrow e$$

is exact.

One will compare this sequence with the exact sequence 1.4, but one will note that a) we have not had to make any hypothesis of flatness, or of separability on the fibers, for $X \rightarrow Y$; b) we have the important complement that the morphism $\pi_1(\bar{X}_0, \bar{a}_0) \rightarrow \pi_1(X, a_0)$ is injective.

This last fact will allow us to compare the fundamental group of the other geometric fibers of X over Y with that of $\bar{X}_0$. Indeed, let y_1 be an arbitrary point of Y , X_1 its fiber and $\bar{X}_1$ its geometric fiber, relative to an algebraically closed extension of $k(y_1)$, $\bar{a}_1$ a geometric point

of $\overline{X}_1$, a_1 its image in X and b_1 its image in Y . Let us choose a “class of paths” from a_1 to a_0 , whence a class of paths from b_1 to b_0 , whence a commutative diagram of homomorphisms:

$$\begin{array}{ccccccc} \pi_1(\overline{X}_1, \overline{a}_1) & \longrightarrow & \pi_1(X, a_1) & \longrightarrow & \pi_1(Y, b_1) & \longrightarrow & e \\ & & \downarrow & & \downarrow & & \\ e & \longrightarrow & \pi_1(\overline{X}_0, \overline{a}_0) & \longrightarrow & \pi_1(X, a_0) & \longrightarrow & \pi_1(Y, b_0) \longrightarrow e, \end{array}$$

where the two vertical arrows drawn are isomorphisms. Since the second row is exact, we therefore find a canonical homomorphism, which we shall call *the specialization homomorphism for the fundamental group* (depending only on the class of paths chosen from a_1 to a_0 , hence *defined modulo inner automorphism of $\pi_1(X, a_0)$*):

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0).$$

When the first row above is also exact, it follows at once that the specialization homomorphism is surjective. We therefore find, taking 1.4 into account:

COROLLARY 2.3. *Under the conditions of 2.1, suppose moreover that the morphism $f: X \rightarrow Y$ is separable (1.1) and $\overline{X}_0$ connected (so that by virtue of 1.2 we have $f_*(\mathcal{O}_X) = \mathcal{O}_Y$). Then for every geometric fiber $\overline{X}_1$ of X over Y , equipped with a geometric point $\overline{a}_1$, the specialization homomorphism defined above is a surjective homomorphism.*

This is a *semicontinuity* result for the fundamental group, which does not yet seem to have an analogue in algebraic topology. One can moreover state it under more general conditions:

COROLLARY 2.4. *Let $f: X \rightarrow Y$ be a proper morphism with geometrically connected fibers, with Y locally noetherian, y_0 and y_1 two points of Y such that $y_0 \in \{y_1\}$, $\overline{X}_0$ and $\overline{X}_1$ the geometric fibers of X corresponding to given algebraically closed extensions of $k(y_0)$ and $k(y_1)$, $\overline{a}_0$ resp. $\overline{a}_1$ a geometric point of $\overline{X}_0$ resp. $\overline{X}_1$. Then one can define in a natural way a specialization homomorphism:*

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0),$$

defined up to inner automorphism, and this is a surjective homomorphism if f is a separable morphism (1.1).

Indeed, it follows first of all from 1.8 that 2.4 is essentially independent of the algebraically closed extensions chosen for the residue fields $k(y_0)$ and $k(y_1)$. This allows us to replace Y by a prescheme Y' over Y having a point y'_0 (resp. y'_1) above y_0 (resp. y_1). We shall then take for Y' the spectrum of the completion of the local ring of y_0 in Y , and we apply 2.3.

REMARKS 2.5. The final conclusion of 2.4 on the surjectivity of the specialization homomorphism, and a fortiori the results 1.3 and 1.4 of which it is a consequence, becomes incorrect if one no longer assumes that $f: X \rightarrow Y$ is a separable morphism, even for projective schemes over an algebraically closed field of characteristic 0. We shall see examples of this later, both in the case where f is flat but where f admits a non-separable fiber (X and Y being nevertheless smooth over k), and in the case where the fibers of f are indeed separable but where f is not flat (for example $f: X \rightarrow Y$ being a birational morphism of integral normal schemes), cf. XI 3. In these examples, it may happen that the fundamental group of the generic geometric fiber is zero, but not that of a suitable special geometric fiber. On the other hand, even if $f: X \rightarrow Y$ is a proper separable morphism as in 2.4, it commonly happens that the specialization morphism is not an isomorphism. Thus, it is easy to give examples where $\overline{X}_1$ is a nonsingular elliptic curve, (hence its fundamental group is commutative, and its ℓ -primary component for a prime number ℓ prime to the characteristic is isomorphic to $\mathbf{Z}_\ell^2$, cf. XI), while $\overline{X}_0$ consists either of two nonsingular rational curves meeting in two points, or of two rational curves tangent at one point, or finally of a rational curve having a singular point which is a cusp. (For the complete classification of degenerate elliptic curves, see the recent work of Kodaira [1] and Néron). One then sees that in these cases, the

fundamental group of $\overline{X}_0$ is respectively $\widehat{\mathbf{Z}}$, e , e , hence “strictly smaller” than that of $\overline{X}_1$. We shall however see later, when f is a *smooth* morphism, a bound for the kernel of the specialization homomorphism, which implies in particular that if $k(y_0)$ is of *characteristic* 0, the specialization homomorphism is an isomorphism. But even for a smooth morphism, if the characteristic of $k(y_0)$ is > 0 , it may happen that the specialization homomorphism is not an isomorphism, as one sees for example in the case where X is an abelian scheme over Y (of relative dimension 1, if one wishes), cf. XI 2. A satisfactory theory of the specialization of the fundamental group must take into account the “continuous component” of the “true” fundamental group, corresponding to the classification of principal coverings with infinitesimal groups as structure group; granting which one would be entitled to expect that the “true” fundamental groups of the geometric fibers of a smooth and proper morphism $f: X \rightarrow Y$ form a nice local system on X , projective limit of finite and flat group schemes over X^2 . We shall return later to this point of view, our present object being on the contrary to push as far as possible the phenomena common to the topological theory and the scheme-theoretic theory of the fundamental group.

Let now X_0 be a proper, smooth and connected curve of genus g over an algebraically closed field k . If k is of characteristic zero, its fundamental group can be determined by transcendental means in the following way. We know that X_0 comes by extension of the base from a curve defined over an algebraically closed extension of finite transcendence degree of the prime field $\mathbf{Q}$, and taking 1.8 into account, we may assume that k is itself of finite transcendence degree over $\mathbf{Q}$. We may therefore assume that k is a subfield of the field $\mathbf{C}$ of complex numbers, and a new application of 1.8 allows us to assume that $k = \mathbf{C}$. It is then not difficult to verify that the fundamental group of X is isomorphic to the compactification of the fundamental group of the associated topological space $\tilde{X}$ (compact oriented surface of genus g), for the topology defined by the subgroups of finite index³. It is on the other hand classical that the topological fundamental group is generated by $2g$ generators s_i, t_i ($1 \leq i \leq g$), subject to a single relation:

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1.$$

Hence the fundamental group of X admits $2g$ *topological* generators s_i, t_i ($1 \leq i \leq g$), bound by the single preceding relation. If now the characteristic of k is $p > 0$, let us denote by A the ring of Witt vectors constructed with k , by K an algebraically closed extension of its field of fractions. We have seen in III 7.4 that there exists a scheme X over $Y = \text{Spec}(A)$, proper and smooth over Y , reducing to X_0 . Applying 2.3 to it, we find a *surjective* morphism

$$\pi_1(X_1) \longrightarrow \pi_1(X_0),$$

where $X_1 = X \otimes_A K$. It is immediate⁴ that X_1 is smooth over K , connected (1.2), of dimension 1, and its genus is equal to g (by the invariance of the Euler-Poincaré characteristic, cf. EGA III 7). Since K is of characteristic 0, one can apply the preceding result to it. We have thus proved by *transcendental means*:

THEOREM 2.6. *Let X_0 be a smooth proper and connected algebraic curve over an algebraically closed field k , and let g be its genus. Then $\pi_1(X_0)$ admits a system of $2g$ topological generators, bound by the relation written above. When the characteristic of k is 0, $\pi_1(X_0)$ is even the free group of Galois type for the generators and the relation which precede.*

REMARKS 2.7. There does not exist at the present time, to the knowledge of the writer, a proof by purely algebraic means of the preceding result, (except for the genera 0, 1). To begin with, one hardly sees how to distinguish in $\pi_1(X_0)$ $2g$ elements, which one could

²This extremely attractive conjecture is unfortunately contradicted by an unpublished example of M. Artin, already when the fibers of f are algebraic curves of given genus $g \geq 2$.

³This deduction was made explicit in one of the oral exposés which were not written up.

⁴cf. EGA IV 12.2.

then expect to form a system of topological generators. In this respect, the situation of the rational line minus n points, and the study of the coverings of it tamely ramified at these points, is more congenial, since the consideration of the ramification groups at these n points furnishes n elements of the fundamental group to be studied, which one indeed shows topologically generate this fundamental group, as we shall see later⁵. But even in this particularly concrete case, there does not seem to exist a purely algebraic proof. Such a proof would obviously be extremely interesting. The only fact concerning the fundamental group of a curve that one knows how to prove by purely algebraic means (with the exception of the weak finiteness theorem 2.12 below, proved by algebraic means by Lang-Serre [2]), seems to be the determination of the abelianized fundamental group via the Jacobian (pointed out in IX 5.8, last line).

2.8. The last assertion of 2.6 is no longer valid in characteristic $p > 0$, as one sees already in the case of elliptic curves. As we have already pointed out, we do not know whether the fundamental group of X_0 is topologically of finite presentation; this seems quite improbable.

THEOREM 2.9. *Let k be an algebraically closed field, and X a proper and connected scheme over k . Then the fundamental group of X is topologically finitely generated.*

We proceed by induction on $n = \dim X$, the assertion being trivial for $n \leq 0$. Suppose therefore $n > 0$, and the theorem proved for the dimensions $n' < n$. By Chow's lemma (EGA II 5.6.2) there exists a projective scheme X' over k and a surjective morphism $X' \rightarrow X$. We may obviously assume X' reduced, and by passing to the normalization, normal. Thanks to the theory of descent, it suffices to prove that the fundamental groups of the connected components of X' are topologically finitely generated (IX 5.2). This therefore brings us back to the case where X is *projective* and *normal*. If then $n = 1$, it suffices to apply 2.6. If $n \geq 2$, we consider a projective immersion $X \rightarrow \mathbf{P}_k^r$, and a hyperplane section $Y = X \cdot H$ (endowed with the induced reduced structure), such that $Y \neq X$, i.e. $H \not\supset X$. We shall then have $\dim Y < n$, and taking into account the induction hypothesis, it suffices to prove that $\pi_1(Y) \rightarrow \pi_1(X)$ is *surjective*. Now more generally:

LEMMA 2.10. *Let X be a prescheme proper over an algebraically closed field k , $g: X \rightarrow \mathbf{P}_k^r$ a morphism. Suppose X irreducible and normal and $\dim g(X) \geq 2$. Let H be a hyperplane of $\mathbf{P}_k^r$ and $Y = X \times_{\mathbf{P}_k^r} H$. Then Y is connected, and the homomorphism $\pi_1(Y) \rightarrow \pi_1(X)$ is surjective.*

These assertions indeed follow from the following one:

COROLLARY 2.11. *Under the preceding conditions, let X' be a connected étale covering of X , and $Y' = X' \times_X Y = X' \times_{\mathbf{P}_k^r} H$ the induced covering on Y . Then Y' is connected.*

Since X is normal, X' is normal, hence, being connected, X' is irreducible; moreover its image in $\mathbf{P}_k^r$ is of dimension ≥ 2 . A well-known lemma due to Zariski (and called "*Bertini's theorem*") therefore implies that if H'_1 is the generic hyperplane in $\mathbf{P}_k^r$, defined over an extension K of k , then $X' \times_{\mathbf{P}^r} H_1$ is universally irreducible hence universally connected over K . Zariski's connectedness theorem (EGA III 4) then implies that for *every* hyperplane H (defined over an arbitrary extension of k) $X' \times_{\mathbf{P}^r} H$ is geometrically connected. This completes the proof of 2.11, hence of 2.9.

COROLLARY 2.12 (Lang-Serre). *Under the conditions of 2.9, for every finite group G , the set of classes, up to isomorphism, of principal coverings of X with group G , is finite.*

REMARK 2.13. Under the conditions of 2.10 we shall prove, when $\dim g(X) \geq 3$ (at least when g is an immersion and X regular), a more precise result, known in algebraic geometry

⁵Cf. Exp. XII. Even so, these elements are really determined only modulo conjugation, and it is appropriate to make a judicious *simultaneous* choice of these elements in their classes.

under the name of “*Lefschetz theorem*”: $\pi_1(Y) \rightarrow \pi_1(X)$ is an isomorphism.⁶ There are, in the classical cases, analogous statements for the homology groups and the higher homotopy groups, which sooner or later will have to be encompassed in abstract algebraic geometry. Even for the Hodge cohomology $H^p(X, \Omega^q)$, it does not seem that the question has yet been studied; it is moreover hardly probable that for this latter, the Lefschetz theorems remain valid as they stand in characteristic $p > 0$.

[Remark 2.14 (added in 2003 (MR)). Let R be a complete discrete valuation ring, with algebraically closed residue field k of characteristic $p > 0$, with field of fractions K , and let Y be a proper and smooth, connected curve of genus g over R . We thus have a surjective specialization morphism $\text{sp} : \pi_1(Y_{\overline{K}}) \rightarrow \pi_1(Y_k)$. We have already noted that if K is of characteristic 0, sp is not an isomorphism as soon as $g \geq 1$. Suppose K of characteristic p , so that R is isomorphic to the ring of formal power series $k[[T]]$. In equal characteristic $p > 0$, the fundamental group is not determined by the genus g , as one already sees on elliptic curves, which may be ordinary or supersingular. Let us cite the recent result of A. Tamagawa (not yet published). If G is a profinite group, we denote by G^{res} the profinite quotient group of G , projective limit of the topological quotients of G which are finite solvable.

THEOREM (A. Tamagawa). *Suppose $g \geq 2$, that the special fiber Y_k is definable over a finite field, and that the morphism $\text{sp}^{\text{res}} : \pi_1(Y_{\overline{K}})^{\text{res}} \rightarrow \pi_1(Y_k)^{\text{res}}$ deduced from sp by passage to the quotient is bijective. Then the curve Y is constant over R .*

Let us note that the Galois group of $\overline{K}/K$ is solvable. The preceding statement can also be translated as follows: suppose that every finite étale covering $X_K \rightarrow Y_K$ of the generic fiber Y_K , Galois with solvable Galois group, has potentially good reduction (that is, extends to a finite étale covering of Y after possible replacement of R by its normalization in a finite extension of K). Then Y is constant over R .]

3. Application of the purity theorem: continuity theorem for the fundamental groups of the fibers of a proper and smooth morphism

Let us recall without proof⁷ the following

PURITY THEOREM 3.1 (Zariski–Nagata). *Let $f: X \rightarrow Y$ be a quasi-finite and dominant morphism of integral preschemes, with X normal, Y regular locally noetherian, and let Z be the set of points of X where f is not étale, i.e. where f is ramified (this amounts to the same thing, I 9.5 (ii)). If $Z \neq X$, Z is of codimension 1 in X at all its points, i.e. for every irreducible component Z' of Z with generic point z , the Krull dimension of $\mathcal{O}_{X,z}$ is equal to 1.*

Recall that a prescheme is said to be *normal* resp. *regular* if its local rings are normal resp. regular, and that the relation $Z \neq X$ also means that the finite extension $R(Z)/R(X)$ (where R denotes the field of rational functions) is *separable*. Placing oneself at the generic point z of a component Z' of Z , and localizing at the point y of Y below z , one finds the equivalent statement:

COROLLARY 3.2. *Let A be a regular noetherian local ring, $A \rightarrow B$ an injective local homomorphism such that B is normal, a localization of an algebra of finite type over A , and quasi-finite over A ; suppose moreover that $\dim A (= \dim B) \geq 2$, and that for every prime ideal $\mathfrak{p}$ of B distinct from the maximal ideal, B is étale over A at $\mathfrak{p}$, i.e. $B_{\mathfrak{p}}$ is étale over $A_{\mathfrak{q}}$ (where $\mathfrak{q} = A \cap \mathfrak{p}$). Then B is étale over A .*

Moreover, it is not difficult to reduce this last statement to the case where A is a *complete* local ring, hence where B is *finite* over A . Zariski [5] gives a simple proof of this result, valid in the case of equal characteristics; the general case is due to Nagata [3], who relies on a delicate result of Chow; the latter has not been checked by any of the participants of the

⁶Cf. the seminar SGA 2 (1962) following this one, theorem X 3.10.

⁷For a proof, cf. SGA 2 X 3.4.

Seminar, and should be the subject of a later exposé. Let us only point out here the very simple proof in the special case where $\dim A = 2$, which is sufficient for the most important application that we shall make of it in the present no. Since B is normal, it is a B -module of depth (old terminology: cohomological codimension) ≥ 2 , hence it is an A -module of depth ≥ 2 , and since A is regular of dimension 2, it follows that B is a *free module* over A ⁸. It then follows from I 4.10 that the set of prime ideals $\mathfrak{q}$ of A at which B is ramified over A is the subset of $\operatorname{Spec}(A)$ defined by a principal ideal (generated by the discriminant of a basis of B over A), hence is empty if it is contained in the closed point of $\operatorname{Spec}(A)$, which proves 3.2 when $\dim A = 2$.

We shall use 3.1 above all in the equivalent form:

COROLLARY 3.3. *Let X be a regular locally noetherian prescheme, U an open subset of X complementary to a closed subset Z of X of codimension ≥ 2 . Then the functor $X' \mapsto X' \times_X U$ from the category of étale coverings of X to the category of étale coverings of U is an equivalence of categories; in particular, if a is a geometric point of U , the canonical homomorphism $\pi_1(U, a) \rightarrow \pi_1(X, a)$ is an isomorphism.*

The last assertion is evidently a consequence of the first, and for the latter one may evidently assume that X is connected, hence irreducible. From the normality of X it already follows that the functor $X' \mapsto X' \times_X U$ from the category of locally free coverings (not necessarily étale) of X' to the category of coverings of U is fully faithful, since the functor $\mathcal{E} \mapsto \mathcal{E}|_U$ from the category of locally free Modules on X to the category of locally free Modules on U is so. It therefore remains to prove that for every étale covering U' of U , there exists an étale covering X' of X (necessarily unique by what precedes), such that U' is isomorphic to $X' \times_X U$. One may evidently assume U' connected, hence irreducible since (U being normal) U' is normal. Let K be the field of rational functions on X , or on U (it is the same), K' that of U' ; then U' is identified with the normalization of U in K' I 10.3. Let X' be the normalization of X in K' (EGA II 6.3); then $X' = X \times_X U \cong U'$; on the other hand X' is normal, integral, and the structural morphism $f: X' \rightarrow X$ is *finite* and dominant (since X is normal and K'/K is a finite separable extension). It is étale in $U' = f^{-1}(U) = X' = f^{-1}(Z)$, and since Z is of codimension ≥ 2 in X , $f^{-1}(Z)$ is of codimension ≥ 2 in X' . One then concludes from 3.1 that X' is étale over X , which completes the proof.

Now let $f: X \rightarrow Y$ be a rational map from a locally noetherian and regular prescheme X to a prescheme Y , and suppose that f is defined in an open set U complementary to a closed subset of codimension ≥ 2 . Then one deduces from 3.3 a functor, defined up to isomorphism, from the category of étale coverings of Y to the category of étale coverings of X , whence for every geometric point a of U , with image b in Y , a canonical homomorphism

$$\pi_1(X, a) \longrightarrow \pi_1(Y, b)$$

(deduced from the canonical homomorphism $\pi_1(U, a) \rightarrow \pi_1(Y, b)$ by means of the isomorphism $\pi_1(U, a) \xrightarrow{\sim} \pi_1(X, a)$). When f is a dominant morphism, X and Y being integral with fields K and L , so that K is an extension of L , and when Y is normal, these correspondences can be made precise in terms of field extensions by noting that for every finite extension L' of L , unramified over Y , the algebra $K' = L' \otimes_L K$ over K is unramified over X .

In particular, these reflections show that the fundamental group of connected regular locally noetherian preschemes, pointed by geometric points localized in codimension ≤ 1 , is a *functor* when one takes as morphisms in this category the dominant rational maps defined in complements of closed subsets of codimension ≥ 2 . Recalling for example that a rational map from a normal scheme over a field k to a scheme proper over k is defined in the complement of a set of codimension ≥ 2 , one finds:

COROLLARY 3.4 (Birational invariance of the fundamental group). *Let k be a field, X and Y two schemes proper over k and regular, $f: X \rightarrow Y$ a birational map from X to Y ,*

⁸Cf. EGA 0_{IV} 17.3.4

Ω an algebraically closed extension of the function field K of X , allowing one to define the fundamental group of X and the fundamental group of Y . The latter are then canonically isomorphic.

This also means that for a finite extension K' of K , if it is unramified over a proper nonsingular “model” X of K , it is so over every other proper nonsingular model.

REMARK 3.5. For other applications of the purity theorem, see the work of ABHYANKAR expounded in [4], inspired by the results of ZARISKI [6, chap. VIII], proved by topological means. The latter are far from having been assimilated by “abstract” algebraic geometry and deserve new efforts.

We shall need a few elementary facts from the theory of ramification. Let V be a discrete valuation ring with field of fractions K , residue field k , L a Galois extension of K with group G , V' the normalization of V in L , which is a free module of rank $n = [L: K]$ over V , $\mathfrak{m}'$ a maximal ideal of V' , G_d the subgroup of G formed by the elements leaving $\mathfrak{m}'$ invariant, so that G_d operates on the residue extension $k' = V'/\mathfrak{m}'$ of k , and G_i the subgroup of the elements of G_d operating trivially (recall that G_d and G_i are called respectively the *decomposition* and *inertia* subgroups of G). One says that L is *tamely ramified* over V if $n_i = [G_i: e]$ is of order p prime to the characteristic of k (a condition always satisfied if k is of characteristic 0). It is well known that G_i then embeds canonically in the group k'^* , hence is isomorphic to the group of n_i th roots of unity in k'^* , which implies in particular that G_i is *cyclic*. The typical case of this situation is the one where one sets $L = K[t]/(t^n - u)$, u being a uniformizer of V and n an integer prime to p : if K contains the n th roots of unity, L is a totally ramified Galois extension of K , with Galois group $G = G_i$ isomorphic to $\mathbf{Z}/n\mathbf{Z}$.

LEMMA 3.6 (ABHYANKAR’S lemma). *Let V be a discrete valuation ring with field of fractions K , L and K' two Galois extensions of K tamely ramified over V , n and m the orders of the corresponding inertia groups, L' a composite extension of L and K' over K . If m is a multiple of n , then L' is unramified over the localizations of the normal closure V' of V in K' .*

Indeed, let W' be the normalization of V' in L' , $\mathfrak{m}'$ a maximal ideal of V' , $\mathfrak{n}'$ a maximal ideal of W' above $\mathfrak{m}'$, $\mathfrak{n}$ the maximal ideal it induces on the normalization W of V in L , G , H , M the Galois groups of L , K' , L' over K , and G_i , H_i , L'_i the inertia groups corresponding to the chosen maximal ideals. Then M embeds in the product $G \times H$ and M_i in the product $G_i \times H_i$, in such a way that the projections $M \rightarrow G$ and $M \rightarrow H$, $M_i \rightarrow G_i$ and $M_i \rightarrow H_i$ are surjective (sorites of the intermediate field). It already follows, since G_i and H_i are by hypothesis cyclic of orders m and n prime to p , that M_i is of order prime to p , hence cyclic, and since m is a multiple of n , so that the elements of $G_i \times H_i$ have zero m th power, M_i is of order dividing m , hence of order equal to m since $M_i \rightarrow H_i$ is surjective. This last homomorphism is therefore also injective. Now its kernel is the inertia group of $\mathfrak{n}'$ above $\mathfrak{m}'$, which proves that L' is unramified over K' at $\mathfrak{n}'$. Whence the lemma.

Let us now place ourselves under the conditions of 2.4, where one has a specialization homomorphism

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0)$$

which is *surjective*, relative to a proper and separable morphism $f: X \rightarrow Y$. We wish to determine more precisely the kernel of this homomorphism. Proceeding as in the proof of 2.4, one sees that in this question one may always assume that Y is the spectrum of a *discrete valuation ring* V , *complete and with algebraically closed residue field* (for one can always find such a ring and a morphism of its spectrum Y' into Y whose image is $\{y_0, y_1\}$). Then one has $X_0 = \overline{X}_0$, $k(y_0) = k = \text{residue field of } V$, $k(y_1) = K = \text{field of fractions of } V$. Let K_s be the separable closure of K , $\overline{K}$ its algebraic closure, and for every subring W of $\overline{K}$ containing V , set $X_W = X \otimes_V W$. In particular one has

$$X_V = X, \quad X_K = X_1, \quad X_{\overline{K}} = \overline{X}_1.$$

Moreover the canonical morphism $\overline{X}_1 = X_{\overline{K}} \rightarrow X_{K_s}$ induces an isomorphism on the fundamental groups (IX 4.11), so that, taking into account the isomorphism 2.1 $\pi_1(X_0) \rightarrow \pi_1(X)$, one is reduced to studying the surjective homomorphism

$$\pi_1(X_{K_s}) \longrightarrow \pi_1(X)$$

associated with the canonical morphism $X_{K_s} \rightarrow X$. The determination of the kernel of the latter amounts to the solution of the following problem: *one has a connected principal covering Z_{K_s} of X_{K_s} , with group G (hence associated with a homomorphism of $\pi_1(X_{K_s})$ into G), determine under what conditions it is isomorphic to the inverse image of a principal covering Z of X with group G .*

Let us note first that K_s is the filtered increasing union of its finite subextensions K' over K , and that consequently Z_{K_s} is isomorphic to the inverse image of a principal covering $Z_{K'}$ of $X_{K'}$ for a suitable K' (one should be careful, however, that for fixed K' , $Z_{K'}$ is not uniquely determined). To say that Z_{K_s} is isomorphic to the inverse image of a principal covering Z of X means that there exists a finite subextension $K'' \supset K'$ of K_s such that $Z_{K''} = Z_{K'} \otimes_{K'} K''$ is isomorphic to $Z \otimes_V K''$. Let us now denote, for a finite subextension K' of K_s , by V' the normalization of V in K' , which is a complete discrete valuation ring with residue field k . Hence the canonical morphism $X_{V'} \rightarrow X_V$ induces an isomorphism for the fibers above the closed points of $Y = \text{Spec}(V)$ and $Y' = \text{Spec}(V')$, and it then follows from 2.1 applied to X_V and $X_{V'}$ that the induced homomorphism on the fundamental groups $\pi_1(X_{V'}) \rightarrow \pi_1(X_V)$ is an isomorphism, or again that every principal covering of $X_{V'}$ is the inverse image of a principal covering of X_V determined up to isomorphism. This therefore implies the following

LEMMA 3.7. *Let $Z_{K'}$ be a connected principal covering of $X_{K'}$ with group G , Z_K its inverse image on X_{K_s} . For the latter to be isomorphic to the inverse image of a principal covering Z of X , it is necessary and sufficient that there exist a finite extension $K'' \supset K'$ of K in K_s such that the principal covering $Z_{K''}$ of $X_{K''}$ is induced by a principal covering of $X_{V''}$.*

Suppose in particular that the $X_{V''}$ are normal (for this it suffices for example that X_0 be normal, and a fortiori that X_0 be simple, cf. I 9.1). Since they are connected, they are therefore irreducible. Let L be the field of rational functions for X and X_K , L' the one for $X_{V'}$ and $X_{K'}$, L'' the one for $X_{V''}$ and $X_{K''}$. Then under the conditions of 3.7, $Z_{K'}$ defines a finite separable extension R' of L' , and $Z_{K''}$ defines the extension $R'' = R' \otimes_{L'} L'' = R' \otimes_{K'} K''$. The condition considered in 3.7 therefore also means that there exists a finite separable extension K'' of K' such that $R'' = R' \otimes_{K'} K''$ is *unramified* over the normal scheme $X_{V''}$ with field $L'' = L' \otimes_{K'} K''$, and not only over the open subset $X_{K''}$ of $X_{V''}$.

We assume from now on that $f: X \rightarrow Y$ is a *smooth* morphism; hence the morphisms $X_{V'} \rightarrow \text{Spec}(V')$ are smooth, hence the schemes $X_{V'}$ are *regular*. Note that the fiber of the closed point of $\text{Spec}(V')$ in $X_{V'}$ is irreducible and of codimension 1. Let $\mathfrak{o}'$ be its local ring, which is therefore a discrete valuation ring with field L' , with residue field isomorphic to the field of rational functions of X_0 , hence having the same characteristic as k . Let us define similarly $\mathfrak{o}''$ in L'' , which is evidently the normalization of $\mathfrak{o}'$ in L'' . It then follows from the purity theorem 3.1 or 3.3 that for R'' to be unramified over $X_{V''}$, it is necessary and sufficient that R'' be unramified over $\mathfrak{o}''$, the normalization of $\mathfrak{o}'$ in L'' .

Let us now note that if u' is a uniformizer of V' , it is also a uniformizer of $\mathfrak{o}'$. If then n is an integer prime to the characteristic p of k , and if one takes $K'' = K'[t]/(t^n - u')$, then K'' is a finite Galois extension of K' and L'' is isomorphic to $L'[t]/(t^n - u')$, hence is tamely ramified over $\mathfrak{o}'$ with inertia group of order n . Suppose then G of order prime to p , which implies that R' is tamely ramified over $\mathfrak{o}'$, and let us take for n a multiple prime to p of the order of the inertia group of R' over $\mathfrak{o}'$ (for example $n = [G: e]$). Applying ABHYANKAR's lemma 3.6, one sees that the condition considered in 3.7 is satisfied.

This proves the following theorem:

THEOREM 3.8. *Let $f: X \rightarrow Y$ be a proper and smooth morphism, with geometrically connected fibers, with Y locally noetherian, y_0 and y_1 two points of Y such that $y_0 \in \overline{\{y_1\}}$, $\overline{X}_0$ and $\overline{X}_1$ the corresponding geometric fibers; consider the specialization homomorphism (2.4) $\pi_1(\overline{X}_1) \rightarrow \pi_1(\overline{X}_0)$. This homomorphism is surjective, and every continuous homomorphism of $\pi_1(\overline{X}_1)$ into a finite group G of order prime to the characteristic p of $k(y_0)$ comes from a homomorphism of $\pi_1(\overline{X}_0)$ into G .*

In other words:

COROLLARY 3.9. *If $k(y_0)$ is of characteristic zero, then the specialization homomorphism is an isomorphism. If $k(y_0)$ is of characteristic $p > 0$, then the kernel of the specialization homomorphism is contained in the intersection of the kernels of the continuous homomorphisms of $\pi_1(\overline{X}_1)$ into finite groups of order prime to p (or again, the closed invariant subgroup generated by a Sylow p -subgroup of the group of Galois type $\pi_1(\overline{X}_1)$); if therefore $\pi_1(\overline{X}_1)^{(p)}$ denotes the quotient group of $\pi_1(\overline{X}_1)$ by the preceding closed subgroup, and if one defines $\pi_1(\overline{X}_0)^{(p)}$ similarly, then the specialization homomorphism induces an isomorphism*

$$\pi_1(\overline{X}_1)^{(p)} \xrightarrow{\sim} \pi_1(\overline{X}_0)^{(p)}$$

One will note that the proof of 3.8 is purely algebraic. Proceeding as in 2.6, one concludes from it *by transcendental means*:

COROLLARY 3.10. *Let X_0 be a proper, smooth and connected curve of genus g over an algebraically closed field of characteristic p . With the notation introduced in 3.9, the group $\pi_1(X_0)^{(p)}$ is isomorphic to $\Gamma^{(p)}$, where Γ is the group of Galois type generated by generators s_i, t_i ($1 \leq i \leq g$) subject to the relation*

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1$$

REMARKS 3.11. In the case where $k(y_0)$ is of characteristic zero, the result 3.9 is well known by transcendental means. One will note that the proof of 3.10 makes use of the purity theorem in the case of unequal characteristics, but only in the case of rings of dimension 2, where the proof of the said theorem is easy and was recalled in the text.

4. Bibliography

- [1] K. KODAIRA, On compact analytic surfaces, Ann. of Math. **71** (1960), p. 111–152.
- [2] S. LANG and J-P. SERRE, Sur les revêtements non ramifiés des variétés algébriques, Amer. J. Math. **79** (1957), p. 319–330.
- [3] M. NAGATA, On the purity of branch loci in regular local rings, Illinois J. Math. **3** (1959), p. 328–333.
- [4] J-P. SERRE, Revêtements ramifiés du plan projectif (d’après S. Abhyankar), Séminaire Bourbaki, May 1960.
- [5] O. ZARISKI, On the purity of the branch locus of algebraic functions, Proc. Nat. Acad. Sc. USA **44** (1958), p. 791–796.
- [6] O. ZARISKI, Algebraic surfaces, Ergebnisse... 1948, Chelsea (New York).